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TripForge

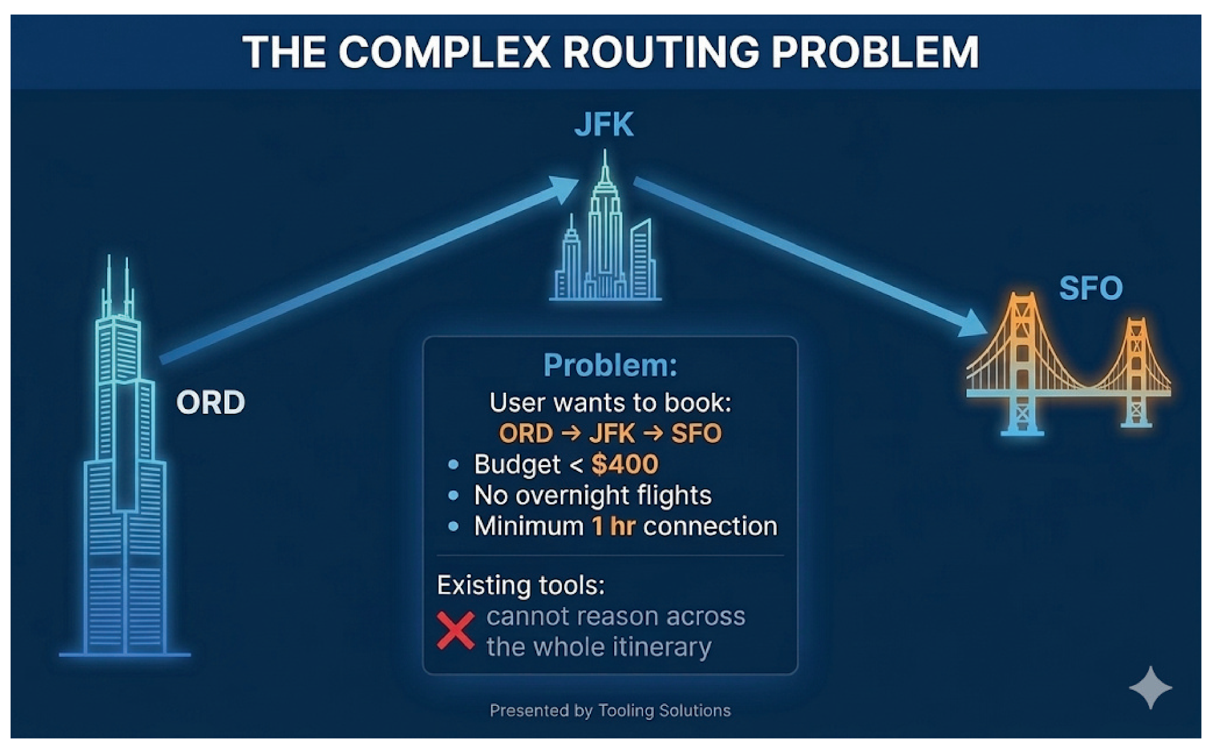
Constraint-Aware Flight Planning

Experience The Demo

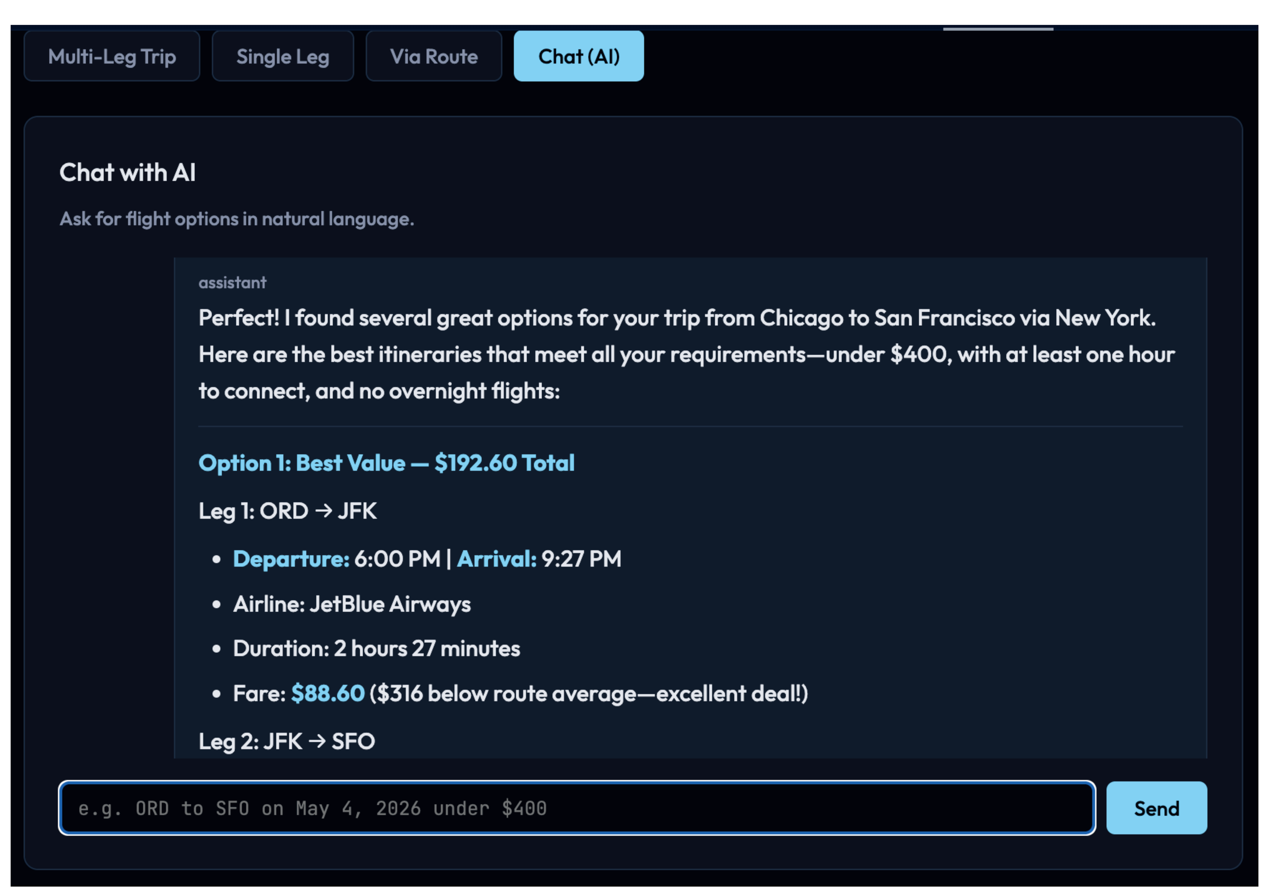
Problem

Planning multi-city trips with constraints is difficult. Travelers must satisfy budgets, connection times, and schedules, while existing tools suggest routes without explicit constraint reasoning.

TripForge models travel as a graph with constraints, enabling reasoning over complete routes and verifiable itineraries.



Demo Output



Why it Matters

Constraint-Aware Planning

Plan multi-city trips with real constraints using reasoning across the full itinerary.

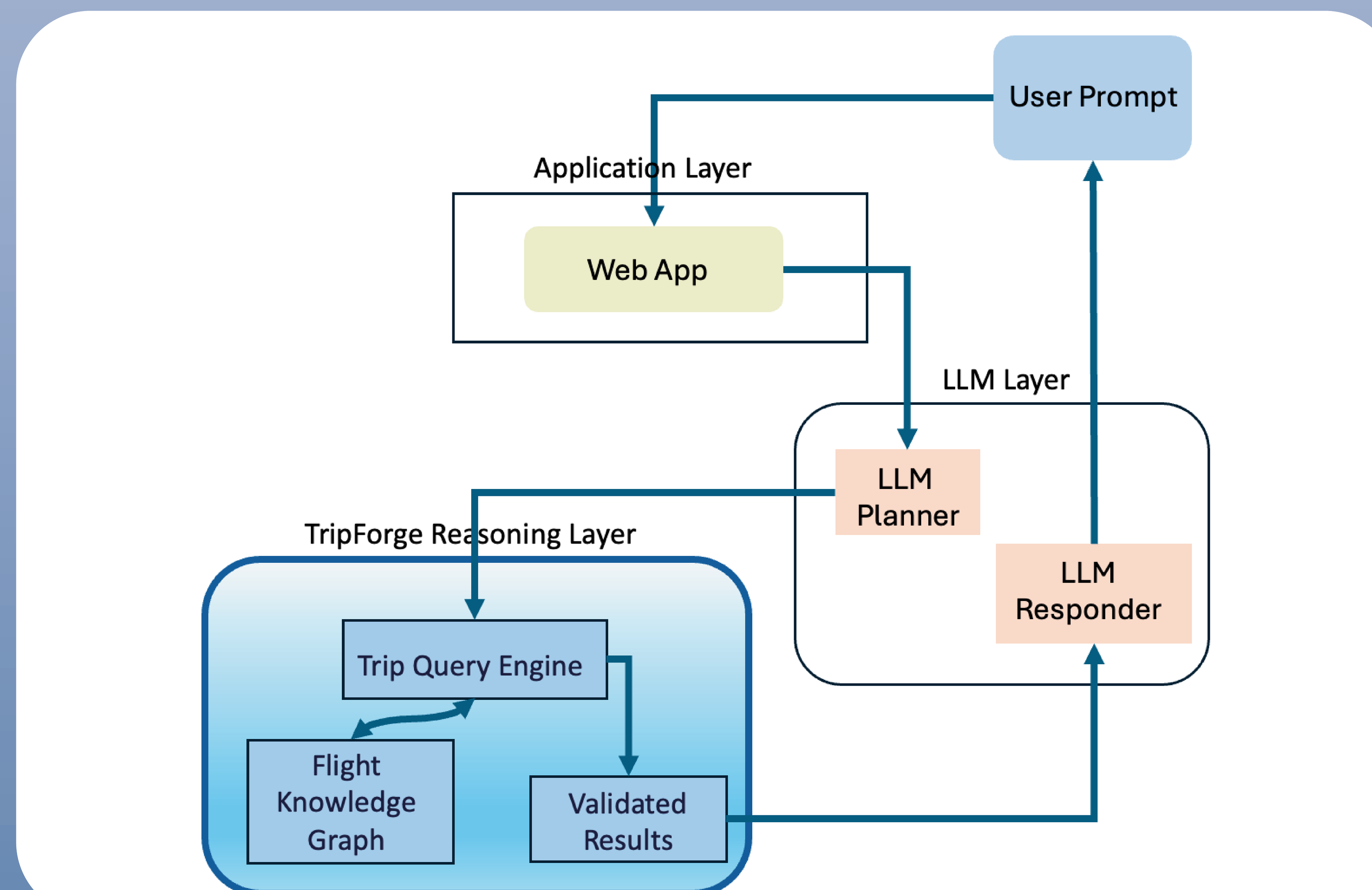
Verifiable Decisions

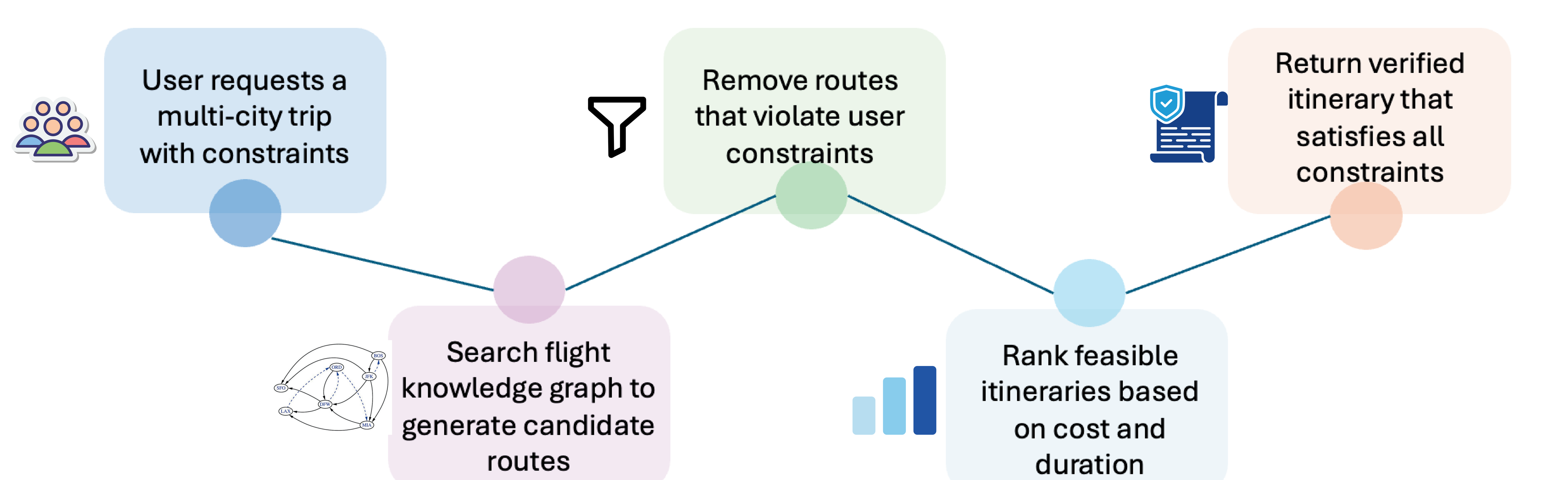
TripForge verifies constraints and reveals trade-offs between cost, duration, and routes.

AI + Deterministic Tools

Ask for trips in plain language and refine plans interactively.

Architecture

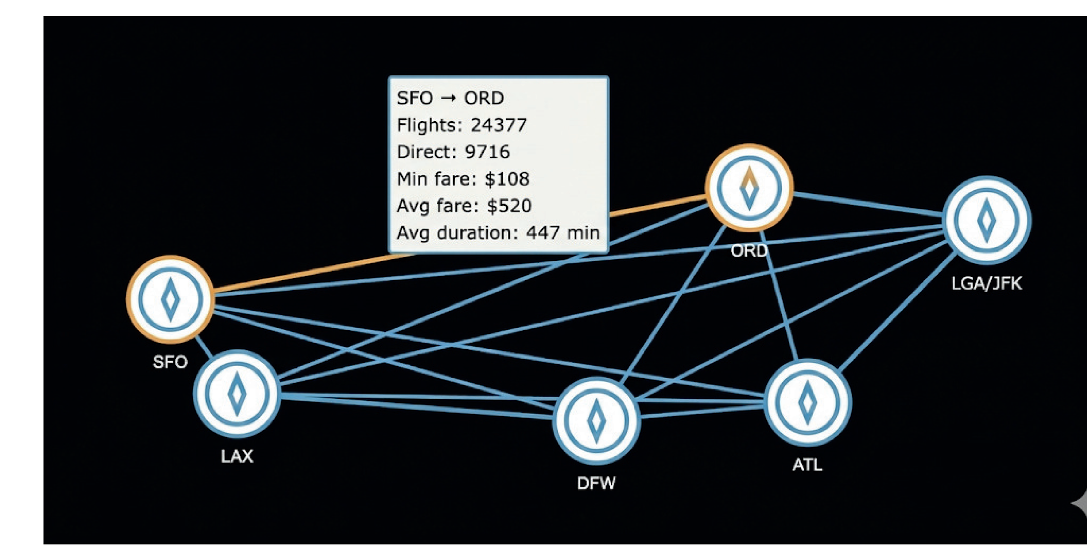




TripForge combines an LLM planning agent with graph-based search and constraint validation to generate feasible itineraries.

Representation

- **Knowledge graph:** airports as nodes, flights as edges.
- **Edges store attributes** such as fare, duration, and departure time.
- **Indexed by origin and weekday** for fast lookup.
- **Supports hard constraints and soft preferences.**



Reasoning & Methods

- **Graph search** explores routes between airports
- **Constraint filtering** enforces budgets and connection times
- **Route scoring** ranks itineraries by cost and duration
- **Pareto analysis** exposes trade-offs between price and travel time
- **Feasibility checks** verify that itineraries satisfy constraints

Future Work

- In-city travel planning (cars, rideshare, transit)
- Full itinerary generation for multi-day trips
- Recommendations for places and activities
- End-to-end holiday planning with constraints